

2: $\int_C F \cdot dr$, $F = \langle x^2, -x^2 \rangle$, C given by $r(t) = \langle t^3, t^2 \rangle$, $0 \leq t \leq 1$

Step 1: Rewrite everything using t .

$$F \cdot dr = F \cdot \frac{dr}{dt} dt = F(r(t)) \cdot \frac{dr}{dt} dt$$

step 2: Substitute for F , find $\frac{dr}{dt}$.

$$r'(t) = \langle 3t^2, 2t \rangle.$$

$$\text{Also, } F(r(t)) = F(t^3, t^2) = \langle t^7, -t^6 \rangle.$$

$$\text{So } \underline{F(r(t)) \cdot \frac{dr}{dt} = 3t^9 - 2t^7}$$

Step 3: Compute integral

$$\int_C F \cdot dr = \int_0^1 (3t^9 - 2t^7) dt = \frac{3}{10} - \frac{2}{8} = \frac{1}{20}.$$

52: Show $F = ((1+xy)e^{xy}, x^2e^{xy})$ conservative & find f with $\nabla f = F$.

Step 1: Recall $f_y = F_2 \Rightarrow f = \int f_y dy = \int F_2 dy$

$$\text{So } f = \int x^2 e^{xy} dy = x e^{xy} + g(x).$$

$$(1+xy)e^{xy} = f_x = e^{xy} + xye^{xy} + g'(x)$$

$$\Rightarrow g'(x) = 0, \text{ so } g = 0 \text{ works, so}$$

$$f = x e^{xy} \text{ works}$$

6: $\int_C F \cdot dr$, C given by $0 \leq t \leq \frac{\pi}{2}$
and $r(t) = \langle \cos t, 2 \sin t \rangle$.

Notice we can apply FTDLI by part 2.

$$\int_C F \cdot dr = \int_C \nabla f \cdot dr = f(\text{end}) - f(\text{start})$$

$$= f(r(\frac{\pi}{2})) - f(r(0)) = f(0, 2) - f(1, 0)$$

$$= 0e^0 - 1e^{0.1} = 0 - 1 = -1.$$

7: $F = \langle ze^y + \cos y, xze^y - x \sin y, xe^y - 2z \rangle$,
find work F does $(0, 3, 2) \rightarrow (1, 0, 2)$.

Problem says use FTOLI, so we need
to find f with $\nabla f = F$.

Step 1: Find this f

$$f = \int f_x dx = \int F_1 dx = \int (ze^y + \cos y) dx \\ = xze^y + x \cos y + g(y, z).$$

$$xze^y - x \sin y + g_y = f_y = F_2 = xze^y - x \sin y$$

$$\Rightarrow g_y = 0 \Rightarrow g(y, z) = h(z).$$

$$xe^y - 2z = f_z = xe^y + 0 + h'(z)$$

$$\Rightarrow h'(z) = -2z \Rightarrow h(z) = -z^2$$

works, so $f = xze^y + x \cos y - z^2$

Step 2: Apply FTOLI

Let C be that smooth path,

Then the required work is

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \int_C \nabla f \cdot d\mathbf{r} = f(\text{end}) - f(\text{start}) \\ &= f(1, 0, 2) - f(0, 3, 2) = \\ &= (2 + 1 - 4) - (0 + 0 - 4) = \boxed{3}.\end{aligned}$$

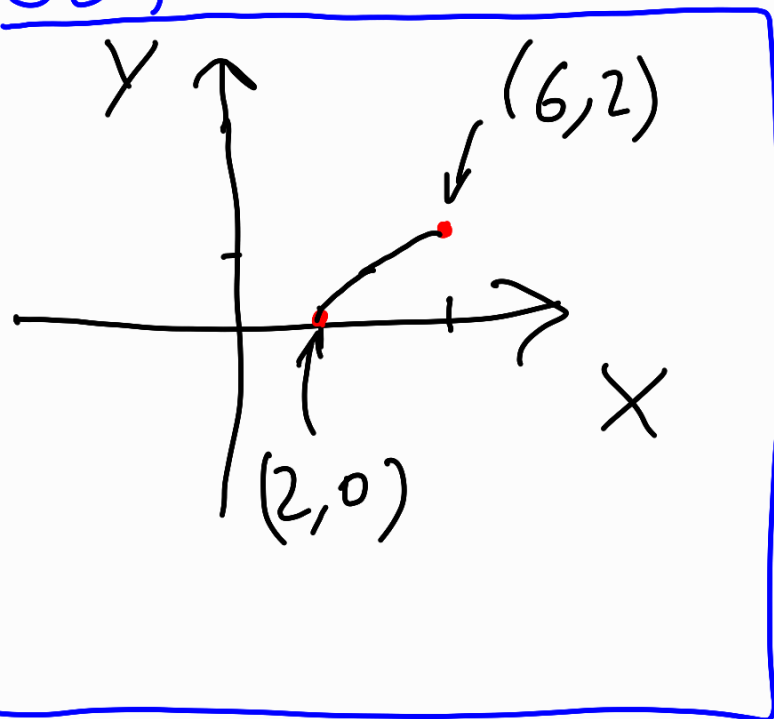
3: Work of $\mathbf{F} = \langle y^2, -y \sin x \rangle$
along $x = y^2 + 2$ from $(2, 0) \rightarrow (6, 2)$
Any work or $\int \mathbf{F} \cdot d\mathbf{r}$ problem
has 2 methods: evaluate integral
manually, or find f with ∇f
 $= \mathbf{F}$ and use FTOLI.

Step 0: Check if latter possible
 $(F_1)_y = (y^2)_y = 2y \neq -y \cos x = (F_2)_x$,
so $\mathbf{F}$ is not conservative &

such f DNE.

So we need to manually evaluate

Step 1: Find curve $r(t)$ & bounds



Set $y = t$,
then $x = y^2 + 2$
 $= t^2 + 2$ and
 $y = 0 \rightarrow 2$ means
 $0 \leq t \leq 2$. So

$$\underline{r(t) = \langle t^2 + 2, t \rangle, 0 \leq t \leq 2}$$

Step 2: Simplify $F \cdot dr$

$$\begin{aligned} F \cdot dr &= F(r(t)) \cdot \frac{dr}{dt} dt = \\ &= F(t^2 + 2, t) \cdot \langle 2t, 1 \rangle dt = \\ &= \langle t^2, -t \sin(t^2 + 2) \rangle \cdot \langle 2t, 1 \rangle dt \\ &= [2t^3 - t \sin(t^2 + 2)] dt \end{aligned}$$

Step 3: Plug in $F \cdot dr$ & eval.

$$W = \int_C F \cdot dr = \int_0^2 (2t^3 - t \sin(t^2 + 2)) dt$$

$$= \left. \frac{t^4}{4} + \frac{1}{2} \cos(t^2 + 2) \right|_0^2 =$$

$$\frac{2^4}{4} + \frac{1}{2} (\cos 6 - \cos 2) =$$

$$4 + \frac{\cos(6) - \cos(2)}{2}$$

Note: You can even use

$$\cos x - \cos y = -2 \sin\left(\frac{x-y}{2}\right) \sin\left(\frac{x+y}{2}\right)$$

to simplify the last fraction
to $-\sin 4 \sin 2$, but this
is strictly optional.

12: $\int_C y dx + 2x dy$ where C is
 $x=3t, y=t^2, 0 \leq t \leq 1$

Step 1: Find dx & dy , substitute

$$dx = \frac{dx}{dt} dt = 3dt$$

$$dy = \frac{dy}{dt} dt = 2t dt$$

$$\begin{aligned} y dx + 2x dy &= [t^2 \cdot 3 + (6t)(2t)] dt \\ &= (3t^2 + 12t^2) dt = 15t^2 dt \end{aligned}$$

Step 2: Eval. integral

$$\text{int} = \int_0^1 15t^2 dt = 5t^3 \Big|_0^1 = 5$$

42: Is $F = \langle xy + y^2, x^2 + 2xy \rangle$

conservative? Find f w/ $\nabla f = F$.

Recall: we must check if

$$(F_1)_y = (F_2)_x, \text{ else } f \text{ DNE}$$

$$(F_1)_y = (xy + y^2)_y = x + 2y$$

$$(F_2)_x = (x^2 + 2xy)_x = 2x + 2y,$$

so $(F_1)_y \neq (F_2)_x$, so F not conservative & f DNE.